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RESEARCH ARTICLE

A Feasibility Study With In-the-Wild Data in Human Interaction Settings: Acoustic-Visual Fusion for Predictive Sound Source Positioning

ZHICHAO CHEN^{1,2}, CHENFEI XU^{1,2}, HUTHAIFA AHMAD², YUYA OKADOME³, HIROSHI ISHIGURO¹, (Member, IEEE), AND YUTAKA NAKAMURA²

¹Graduate School of Engineering Science, Osaka University, Osaka 565-0871, Japan

²RIKEN Information Research and Development and Strategy Headquarters, RIKEN, Kyoto 619-0288, Japan

³Department of Information and Computer Technology, Tokyo University of Science, Tokyo 125-8585, Japan

Corresponding author: Zhichao Chen (chen.zhichao@irl.sys.es.osaka-u.ac.jp)

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ABSTRACT Acoustic data inherently contains spatial information regarding sounds. Most robotic systems rely primarily on speech recognition results for control; however, the utilization of rich directional cues could enhance their functionality. Especially in human-robot interaction, both verbal content and spatial context in acoustic data are crucial. To capture such auditory context, we employ acoustic maps in this research. These maps represent soundscapes as 2D projections, allowing robots to integrate auditory spatial cues with visual perception. Using a custom acoustic-visual module, we collect egocentric data from four daily scenarios and generate corresponding acoustic maps. By demonstrating a predictive sound source positioning task, we confirm the potential of acoustic maps to capture dialogue-relevant information. Based on these findings, we anticipate potential applications for robots, such as identifying individuals attempting to engage with the robot in crowded environments or predicting turn-taking in multi-party conversations.

INDEX TERMS Human-robot interaction, acoustic mapping, predictive sound source positioning, multimodality learning.

I. INTRODUCTION

Acoustic data processing plays a critical role in real-world human-robot interaction (HRI). Most robotic systems rely on speech recognition to extract verbal content and interpret human audio inputs [1], [2], [3]. Although state-of-the-art methods have significantly enhanced robustness, processing audio in uncontrolled environments remains a considerable challenge [4]. To circumvent these difficulties, some studies have explored vision-only interaction strategies, enabling robots to engage effectively in acoustically challenging settings such as crowded malls [5]. However, other acoustic characteristics, especially spatial information embedded in sound, remain underexplored in comparison to verbal content.

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Acoustic data inherently contains spatial cues essential for contextual comprehension. By integrating beamformed audio signals with visual perception, a visually interpretable modality known as acoustic maps is produced [6]. These maps represent soundscapes as 2D projections, offering an intuitive view for context-aware HRI. Nevertheless, robots with limited microphones use beamforming primarily for noise filtering, overlooking rich spatial details [7]. Moreover, current approaches that exploit spatial audio cues remain confined to single-source identification and neglect fine-grained spatial details [8], [9], [10], [11], [12].

To fully leverage the spatial information in sounds, we propose employing acoustic maps as an alternative visual modality for better robotic perception. To validate this concept, we develop a customized acoustic-visual module, designed to capture egocentric data across four in-the-wild scenarios. Through systematic evaluation of classic

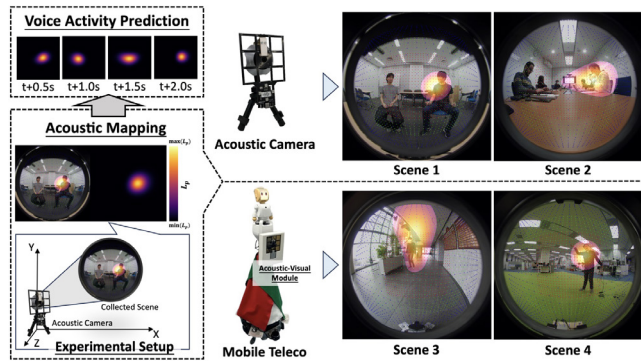


FIGURE 1. (Lower left) Experiment setup: Scenes 1-2 use a fixed acoustic camera; Scenes 3-4 mount it on a teleoperated Mobile Teleco. (Upper left) DAS beamforming at 2 kHz (1/3-octave) visualized with an inferno colormap for PSSP. (Right) Daily scenarios: (1) chat, (2) lab, (3) lobby, (4) office.

beamformers and meticulous grid configurations, we identify the optimal settings for producing high-fidelity acoustic maps. Our system supports a multimodal spatio-temporal prediction task, Predictive Sound Source Positioning (PSSP), in which the auditory spatial distribution is used as an indicator to predict dialogue-relevant events. The task demonstrates not only the feasibility of our approach but also its potential to capture and extract critical features necessary for interactive communication in noisy environments.

The main contributions of this work are as follows:

- **Acoustic-Visual Sensing System:** A synchronized system captures RGB video and directional audio to generate acoustic maps for robotic perception.
- **PSSP and Metric:** PSSP as a novel task with Peak Success Rate, a spatially aware metric complementing MSE, and present an acoustic-visual fusion model for PSSP.
- **In-the-Wild HRI Dataset:** A 436-minute multimodal dataset of 82 clips across four daily interaction scenes, featuring multi-speaker presence and spatial dynamics.
- **Comprehensive Benchmarking:** Extensive evaluation of input-output spatio-temporal configurations for PSSP.

The remainder of this paper is organized as follows: Section III presents the acoustic-visual module design and surveys related work on acoustic maps in robotic systems; Section IV details the generation of high-fidelity acoustic maps, covering beamforming methods, frequency selection, and intensity transformations; Section V introduces the PSSP task and the spatio-temporal prediction model; Section VI reports experiments on acoustic representation learning and multimodal integration; and Section VII discusses the implications of our findings and outlines future directions for adaptive auditory attention in HRI.

II. RELATED WORK

As summarized in Table 1, prior work on voice activity detection (VAD), voice prediction, and voice activity projection (VAP) focuses on detecting speech or identifying active speakers, without modeling spatial dynamics. In contrast,

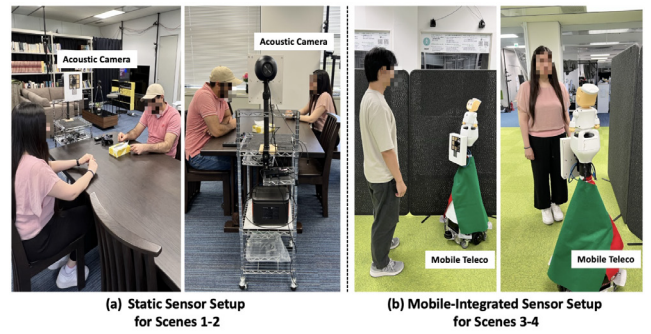


FIGURE 2. Third-person perspectives of the sensor configurations during data collection.

our PSSP formulation predicts future 2D sound maps from past audio and video, capturing spatio-temporal patterns in dynamic, multi-speaker settings. To our knowledge, PSSP is the first to jointly predict spatial sound and its temporal evolution from multimodal inputs, enabling future-aware audio-visual perception. A detailed discussion of related work is provided in the following subsection.

A. DOA-BASED SPATIAL AUDIO IN ROBOTICS

Most robotic hearing systems reduce acoustic scenes to direction-of-arrival (DoA) estimates using features like waveforms, log-Mel spectrograms, or MFCCs. Dual-microphone DoA systems enable active speaker localization in azimuth and elevation [8], [9], and have supported tasks like talker-following and tour-guiding [10], [11]. Visual cues further aid DoA in teleoperation [12] and improve VAD via spatial audio-visual integration [13], [14]. However, DoA offers only coarse spatial resolution, ignoring fine-grained patterns.

B. SPEECH ACTIVITY FORECASTING

Predictive voice activity modeling goes beyond detection to forecasting future audio and turn-taking. Methods like DiffWave [16] and WaveNet [15] predict future STFT frames or raw audio. VAP predicts “who” speaks next using dialogue-based self-supervision [17], [18] or multimodal encoders [19], and its lack of spatial awareness limits its application to two-person dialogues.

Unlike this, PSSP forecasts both the location and timing of sound events by regressing dense spatio-temporal acoustic maps. Although not explicitly designed for turn-taking prediction, its independence from subject placement and specialized equipment (e.g., lapel microphones) makes it well-suited for deployment in unconstrained environments. By adopting the naive yet practical assumption that sound sources localized near a person’s head correspond to speech, the system enables straightforward speech prediction. This study thus investigates a multimodal sensor-based system for future speaker prediction, aiming to establish a core function for conversational robots in multi-party settings.

TABLE 1. Comparison of PSSP formulation with prior audio-visual prediction methods.

Task	Input Modality	Output Target	Future Prediction	Objective
VAD [8]–[11]	Waveform / log-Mel / MFCC	Binary Speech Activity	No	Detect the presence or absence of speech in the current frame
Audio-Visual VAD [12]–[14]	Audio + Video (Face / Mouth Region)	Binary Speech Activity	No	Enhance VAD performance by incorporating visual cues
Voice Prediction [15], [16]	Past Audio (Waveform or STFT Frames)	Future Audio Waveform	Yes (Regression)	Predict upcoming speech signal based on previous audio context
VAP [17]–[19]	Audio + Video	Next Active Speaker Identity	Yes (Classification)	Anticipate which participant will speak next
PSSP (ours)	Sequential Sound Maps + Video	Future Sound Maps	Yes (Regression)	Predict the spatio-temporal evolution of sound dynamics

C. ACOUSTIC MAP REPRESENTATIONS AND PSSP

Acoustic maps capture pixel-level sound intensity, offering rich spatial context for robotic perception. Building on advances in acoustic imaging [6], an acoustic-visual system is developed to generate these maps. Yet, their temporal dynamics remain underexplored. To address this, we propose PSSP—the first approach to jointly predict future sound maps and their temporal dynamics from multimodal inputs. PSSP distinguishes itself from prior works by enabling spatio-temporal prediction, capturing the when and where of upcoming vocal events.

D. MULTIMODAL VIDEO PREDICTION

To realize PSSP, we adopt a CNN-based video prediction architecture that fuses audio and visual cues. While RNNs and Transformers have advanced video prediction [20], [21], CNNs offer competitive performance with better efficiency [22]. Our approach aligns with multimodal learning trends, where joint spatio-temporal reasoning aids tasks like speech separation [23] and speaker diarization [24]. However, the role of visual context in predictive spatio-temporal audio modeling remains underexplored. This work addresses that gap through explicit modeling.

III. ACOUSTIC CAMERA

To enhance the use of acoustic data in robotic perception, we exploit auditory spatial cues. A microphone array assembled with a camera forms an acoustic camera, producing acoustic maps that align sound maps with visual data. These maps synergize auditory dynamics with visual context, laying the groundwork for more perceptually aware robotic systems.

A. AUDIOVISUAL SITUATIONAL AWARENESS SENSOR

The acoustic camera features a webcam fitted with a 180° circumferential fisheye lens,¹ capable of capturing

full-HD RGB images at 30 frames per second. It also includes a 16-channel microphone array² arranged in a square configuration with uniform spacing of 42 mm and a sampling rate of 44.1 kHz. To enable accurate acoustic mapping, the camera lens is precisely aligned with the central opening of the microphone array. Figure 1 presents the sensor system architecture, detailing its inputs and outputs. In some later experimental setups, an acoustic camera was additionally integrated as an external module into Mobile Teleco, a mobile variant of the conversation robot CommU.³ In these cases, the robot was employed as a remotely operated interactive avatar.

IV. ACOUSTIC MAP EXTRACTION

To extract acoustic maps, we follow the four-stage approach in Acoular [6] with customized settings: (1) cross-spectral matrix (CSM) construction, (2) grid definition, (3) beam-forming, and (4) intensity transformation. Acoustic map generation on a standard CPU setup requires approximately 0.2 seconds per frame. The key parameters involved in each stage are summarized in Table 2, with detailed descriptions provided in the following subsections.

A. CSM CONSTRUCTION

To construct the CSM, signals are framed with overlap and windowed before computing the cross-power spectra $S_{i,j}$ between microphone pairs i and j . The window function is critical for reducing spectral leakage and enhancing frequency resolution. Typical options include the Hanning, Hamming, and Blackman-Harris windows [25], [26], [27]. For acoustic mapping, we prefer the Blackman-Harris window for its superior sidelobe suppression ability, which is essential for precise localization in multi-source environments.

A 66.1% overlap is ideal for the Blackman-Harris window, whereas 50% and 75% are commonly used for others [27].

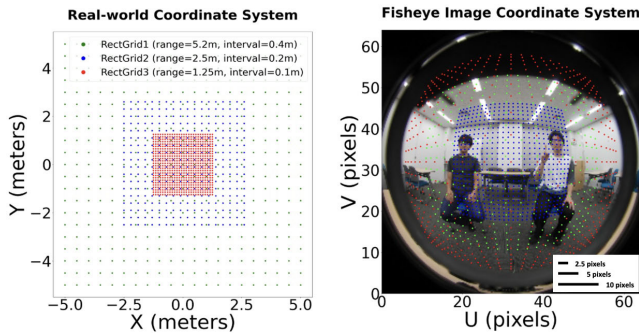
¹Xacti Fisheye Camera: <https://xacti-co.com>

²UMA-16 v2 USB Mic Array: <https://www.minidsp.com/products>

³CommU Robot: <https://www.vstone.co.jp/products/commu/spec.html>

TABLE 2. Critical parameters for acoustic map generation.

Parameter	Value / Description
Audio Device	16 channels at 44.1 kHz sampling rate
Audio Length	20480 samples
Window Function	Blackman-Harris Window (66.1% overlap)
Window Size	4096 samples
Grid Configuration	Composite grid strategy to maximize spatial information (see Section IV-B)
Beamforming	Functional Beamformer within 2–8 kHz
Intensity Transform	Enhancement of dominant sound sources (see Section IV-D)

**FIGURE 3.** (Left) Grid points in the real-world coordinate system. (Right) Projections on a 64 × 64 fisheye image with scale bars.

It ensures a constant sum of windowed segments, thereby minimizing amplitude modulation and preserving spectral fidelity. Moreover, the choice of window size and input length involves a trade-off between frequency detail and computational cost. Following [28], a 4096-sample window (~ 10.8 Hz resolution at 44.1 kHz) and 20480-sample input (~ 0.465 s) are employed, ensuring adequate temporal context for capturing major spectral features.

B. GRID DEFINITION

The grid serves as the spatial basis for computing acoustic energy. We uniformly sample points on a 2D plane 1.5 m in front of the device. Due to the fisheye distortion, projections appear dense near the edges and sparse at the center. To ensure sufficient spatial information, we merge three square grids with distinct configurations into one. In Figure 3, Grid 1 (10.4 m side, 0.4 m interval), Grid 2 (5 m, 0.2 m), and Grid 3 (2.5 m, 0.1 m) contain 729, 676, and 676 points, respectively. The unified grid consists of 2081 non-overlapping points.

To align with the fisheye image, a real-world point $P_w = (X_w, Y_w, Z_w)$ is projected. In perspective projection, the radial distance $r = f \cdot \tan(\theta)$, where f is the focal length and $\theta = \arctan(\frac{\sqrt{X_w^2 + Y_w^2}}{Z_w})$ is the angle of incidence [29]. Due to the fisheye radial distortion, we adopt the equidistant projection model that $r = f \cdot \theta$ where maximum r is half the image

resolution L . With the azimuthal angle $\varphi = \arctan(\frac{Y_w}{X_w})$, P_w is projected to fisheye polar coordinates:

$$(r, \varphi) = (f \cdot \arctan(\sqrt{X_w^2 + Y_w^2}/Z_w), \arctan(Y_w/X_w)) \quad (1)$$

We then normalize r as $r' = \arctan(\sqrt{X_w^2 + Y_w^2}/Z_w) \cdot L/\pi$, and convert to Cartesian coordinates:

$$u = L/2 + r' \cdot \cos \varphi, \quad v = L/2 + r' \cdot \sin \varphi \quad (2)$$

This conversion offers precise mapping from the real-world coordinates to the fisheye image Cartesian coordinate system.

C. BEAMFORMING

The beamformer estimates acoustic power at grid points, producing spatial maps for sound source localization.

1) STEERING VECTOR COMPUTATION

A steering vector $\omega(x)$ models the expected phase delays across the microphone array for a wavefront emanating from a grid point x .

2) BEAMFORMER SELECTION

In Figure 4 (Left), we compare four frequency-domain beamformers: delay-and-sum (DAS), functional, orthogonal, and Capon beamformers [30], [31], [32], [33]. The functional beamformer is adopted for its superior ability to sharpen sound sources. During beamforming, the sound pressure is derived from $P(x) = (\frac{\omega^H(x)S\omega(x)}{tr(S)})q$, where $\omega^H(x)$ is the conjugate transpose of the steering vector, $tr(S)$ is the trace of the CSM, and q controls peak sharpness and sidelobe suppression. The resulting $P(x)$ forms an acoustic map, with peaks marking potential source locations.

3) FREQUENCY BAND

Frequency band influences the spatial resolution. High-frequency offers fine spatial detail but is sensitive to noise, whereas low-frequency provides wide coverage with reduced resolution. Audio devices typically operate between 200 Hz and 10 kHz to raise intelligibility and suppress noise [34]. Furthermore, spatial aliasing may occur above the Nyquist frequency $f_N \approx 4107\text{Hz}$ for a microphone spacing of 42 mm [30]. Advanced beamformers mitigate aliasing effects, enabling reliable use of higher frequencies. Hence, we assess six candidates in Table 4 (Right): (1) 2000 Hz (1/3-octave: 1782–2245 Hz), (2) 8000 Hz (1/3-octave: 7120–8980 Hz), (3) 2000–4000 Hz, (4) 2000–8000 Hz, (5) 1500–8500 Hz, and (6) 200–10000 Hz. The 2000–8000 Hz band yields the most stable and refined outcomes.

D. INTENSITY TRANSFORMATION

The beamformer outputs a map of estimated squared sound pressure levels p^2 in units of Pa^2 . These values are converted to sound pressure levels in decibels (dB) using $L_p = 10 \log_{10}(p^2/p_{\text{ref}}^2)$, $L_p \in \mathbb{R}^{H \times W}$, where the reference pressure $p_{\text{ref}} = 20\mu\text{Pa}$. While L_p generally ranges from 0 to 160 dB, it falls between 90 and 120 dB for sound sources

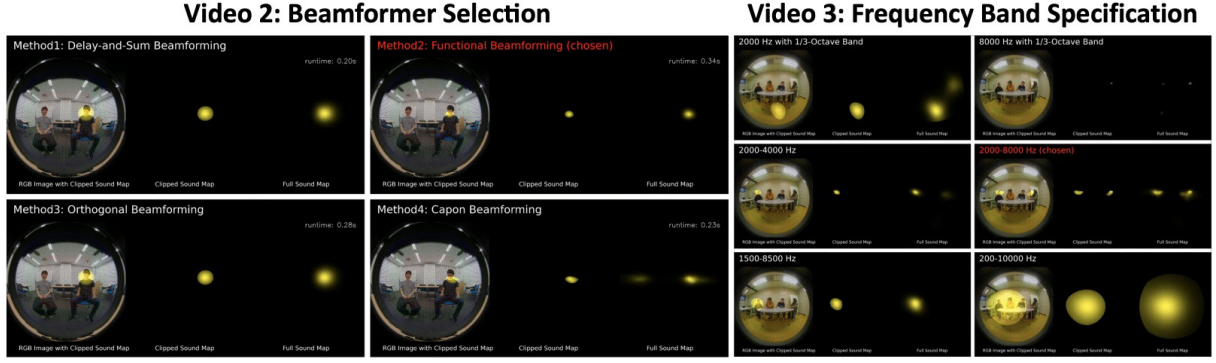


FIGURE 4. (Left) Video 2: Comparison of beamformers; (Right) Video 3: Comparison of frequency bands in scene 10.

in our application. To enhance contrast, we convert $L_p(i, j)$ to $a(i, j)$ under three transformations. “Exponential Scaling by Maximum” is eventually adopted for its superior peak enhancement and effective suppression of irrelevant regions:

- 1) Min-Max Normalization: $a(i, j) = L_p(i, j)/160$.
- 2) Exponential with Min-Max Norm: $a(i, j) = \frac{e^{L_p(i, j)} - e^0}{e^{160} - e^0}$.
- 3) Exponential Scaling by Max: $a(i, j) = e^{L_p(i, j) - \max(L_p)}$.

V. PREDICTIVE SOUND SOURCE POSITIONING

This study aims to enhance robots’ recognition by incorporating spatial information of sounds. As a first step, we develop an acoustic map acquisition system and conduct a pilot experiment to assess whether dialogue-relevant knowledge can be captured. To this end, we adopt a spatio-temporal prediction task that models essential dependencies for predicting future states. This approach exploits the data’s inherent temporal structure, obviating the need for manual labeling. This is a notable advantage given the high cost of annotation in HRI contexts.

A. TASK DESCRIPTION

PSSP involves predicting sound maps from past observations, which is challenging with visual cues alone. Our formulation is inspired by the voice activity projection problem, in which future binary voice-activity flags at +0.2, +0.6, +1.2, and +2.0 seconds are forecasted from raw audio [17]. Here, we define PSSP as the task of inferring future sound source regions with elevated intensities from predicted sound maps. While a PSSP model estimates the full spatial distribution of acoustic dynamics, evaluation focuses on the accuracy of the predicted source areas. We make no claim that this naive heuristic fully captures real-world voice activity dynamics; rather, we adopt it as a minimal yet effective criterion for validating PSSP performance. To avoid redundancy and ensure temporal consistency, we adopt a 0.5-second sampling interval and use ten past states for prediction at +0.5, +1.0, +1.5, and +2.0 s. By incorporating acoustic maps as visual-like inputs alongside other visual modalities, we sidestep the need for intricate multimodal learning that typically demands sizable paired datasets.

To model this, the acoustic map $I_{AM}(t | \alpha) \in \mathbb{R}^{1 \times C_0 \times H \times W}$ is defined as a linear fusion of the sound map $I_{SM}(t)$ with transformed values $a(i, j)$ and the corresponding grayscale image $I_{Gray}(t)$ at time t , where $\alpha \in [0, 1]$ is the fusion ratio:

$$I_{AM}(t | \alpha) = \alpha \cdot I_{SM}(t) + (1 - \alpha) \cdot I_{Gray}(t) \quad (3)$$

The additional visual input $I_{Mods}(t) \in \mathbb{R}^{1 \times C_m \times H \times W}$ is formed by channel-wise concatenation of individual representations $I_{m_i}(t) = F_i(I_{RGB}(t)) \in \mathbb{R}^{1 \times C_i \times H \times W}$ at time t , where a total of N_{vis} modality-specific functions F_i are applied to the RGB image (see Section V-B1) and $\sum_{i=1}^{N_{vis}} C_i = C_m$.

$$I_{Mods}(t) = \text{Concat}_C(\{I_{m_i}(t) | i \in [1, N_{vis}]\}) \quad (4)$$

Given sequences of acoustic maps $A(t | \alpha) = \{I_{AM}(\tau | \alpha) | \tau \in (t - t_h, t]\}$ and visual inputs $M(t) = \{I_{Mods}(\tau) | \tau \in (t - t_h, t]\}$, the prediction model $G(\cdot)$ takes the combined input $X(t) \in \mathbb{R}^{t_h \times C \times H \times W}$, where $C_0 + C_m = C$, to estimate future sound maps $\hat{Y}(t) \in \mathbb{R}^{t_p \times C \times H \times W}$ by minimizing the Mean Squared Error (MSE) loss using ground truth $Y(t) = \{I_{SM}(\tau)\}_{\tau=t+1}^{t+t_p}$:

$$G: X(t) = \text{Concat}_C(A, M) \rightarrow \hat{Y}(t) = \{\hat{I}_{SM}(\tau)\}_{\tau=t+1}^{t+t_p} \quad (5)$$

$$\mathcal{L}_{MSE} = \|Y(t) - \hat{Y}(t)\|_2^2 \quad (6)$$

For better performance, we explore various model configurations by altering input–output mappings in the form $G_{\mathcal{I} \rightarrow \mathcal{O}}: X_{\mathcal{I}}(t) \rightarrow \hat{Y}_{\mathcal{O}}(t)$, where $\mathcal{I} \in \{A, A+M\}$ denotes the input type, varying with the choice of α and visual modalities in I_{Mods} , and $\mathcal{O} \in \{SM, AM, SM+Mods\}$ determines the prediction target. Specifically, Equation 5 corresponds to the case $G_{A+M \rightarrow SM}$ with $\mathcal{I} = A+M$ and $\mathcal{O} = SM$.

For evaluation, Peak Distance $D_{\text{peak}} = \|p_1 - p_2\|_2$ measures the Euclidean distance between p_1 and p_2 , the coordinates of maximum intensities in the prediction $\hat{I}_{SM}(\tau)$ and ground truth $I_{SM}(\tau)$. For robustness, raw sound maps are first smoothed using a local mean filter. For a pixel (i, j) in I_{SM} , the filtered value is computed as the average of intensities within a $k \times k$ neighborhood centered at (i, j) :

$$I_{SM}^{(k)}(i, j) = \frac{1}{k^2} \sum_{x_1=i-\frac{k-1}{2}}^{i+\frac{k-1}{2}} \sum_{x_2=j-\frac{k-1}{2}}^{j+\frac{k-1}{2}} I_{SM}(x_1, x_2) \quad (7)$$

When $k = 1$, the filter has no effect, corresponding to the original sound map. This filtering yields more robust peak distance measurements than direct comparison on raw maps.

To derive a more informative metric, we define Peak Success Rate (PSR) as the percentage of D_{peak} below a threshold. Given the right-skewed, gamma-like distribution of peak distances, PSR effectively reflects how often a PSSP model predicts accurately. Formally, for threshold n , kernel size k , and N samples, PSR is defined as:

$$PSR_k @ n = 1/N \cdot \sum_{i=1}^N I_{SM}^{(k)}(D_{\text{peak}}^{(k)} < n) \quad (8)$$

B. MULTIMODAL SPATIO-TEMPORAL PREDICTION

To investigate the feasibility of retrieving dialogue-relevant information from acoustic maps, we adopt a multimodal prediction model for PSSP, as outlined in Figure 5.

1) MULTIMODALITY

We examine multimodal learning by integrating commonly used visual modalities I_{Mods} , all extracted by TaskPrompter [35]: semantic segmentation⁴ (SemSeg), boundary detection (Edge), surface normal estimation (Normals), Depth, Human Parsing⁵ (Human Parts), Saliency (Sal). Furthermore, we use YOLOv8 to detect faces [36], of which the head poses and emotional states are detected using 6DRepNet360 [37] and EmoNet [38]. Eye and mouth openness are derived from facial landmarks obtained via STAR [39], providing indicators of facial activity. All modalities⁶ are initially extracted from 1080×1080 RGB images and resized to 64×64 as model inputs.

2) MODEL ARCHITECTURE

To validate the use of acoustic maps in robotic perception, we target the PSSP task to learn spatio-temporal patterns from sound map sequences. Rather than pursuing state-of-the-art performance, we aim to confirm the effective capture of spatial and temporal cues. Given the small-scale dataset and real-world deployment goals, we prioritize a lightweight CNN-based architecture over RNNs and Transformers, balancing efficiency and accuracy per prior work [22].

In practice, our model architecture is built upon the CNN translator module from SimVP [22]. To enhance learning efficiency, input sequences are first encoded into the latent space via a four-layer CNN encoder. The translated latent features are then decoded by a five-layer CNN decoder. Each layer is followed by GroupNorm and a SiLU activation. Random shifting is applied as data augmentation. We use a batch size of 32, a learning rate of 1×10^{-6} , and the AdamW optimizer with a weight decay of 0.01. 90% of the data is used for training. Although the architecture allows equal-length

input and output sequences, we report only the first four prediction steps due to task difficulty.

3) TEXTURE AUGMENTATION

Early layers of CNNs primarily capture low-level features such as edges and textures [40]. However, sound maps exhibit sparse textures and blurred edges, which may hinder the learning of discriminative features. To address this, we explore whether acoustic maps can enhance texture information and improve feature extraction. To preserve the dominance of acoustic cues, we use $\alpha \in \{0.5, 0.7, 0.9, 1.0\}$ with Equation 3. When $\alpha = 1$, fusion is disabled and I_{AM} equals I_{SM} . The effect of varying α is evaluated in SM-only and full-modality prediction settings, with results reported in Sections VI-A and VI-D.

C. IN-THE-WILD DATASET

While the acoustic data is collected via Zoom meeting in VAP settings [17], [18], our acoustic-visual data is gathered in four scenarios of real-world human interaction settings. For clarity, we label Scenes 1-4 as Chat, MTG, 1F, 3F, and categorize them into two types: 1) Fixed talk scene (Chat, MTG) captures continuous dialogue with a fixed device; 2) Mobile noise scene (1F, 3F) features minimal human activity and environmental noise recorded by a moving Mobile Teleco. The third-person views of the sensor configurations during data collection are illustrated in Figure 2.

- Scene 1 (Chat): Three students form three 2-person pairs and engage in a chat. The experiment produces 96,031 samples from three 18-minute video clips.
- Scene 2 (MTG): Meetings with student presentations and professor feedback (third person from left). The recording has 125,220 samples from two 35-min videos.
- Scene 3 (1F): Mobile Teleco is teleoperated in a lobby with passing pedestrians. It contains 315,851 samples over 2.9 hours from 49 video clips (1-10 mins)
- Scene 4 (3F): Mobile Teleco is teleoperated in an office where staff work. It includes 250,930 samples over 2.3 hours from 28 video clips (2-9 mins).

VI. EXPERIMENTS

To optimize model configurations for PSSP, we conduct controlled ablation studies examining key factors affecting prediction accuracy and metric reliability. These include PSR evaluation, modality configuration, and output representation. Each study isolates a specific design variable to identify optimal settings and assess performance impact.

Table 3 summarizes the key findings from experiments 1-5. Exp.1 reveals a temporal decline in prediction accuracy, supporting the use of Aggregate PSR. Exp.2 shows that PSR is sensitive to kernel sizes and thresholds, with optimal values empirically determined. Exp.3 demonstrates original sound maps outperform fused variants, while Exp.4 confirms that grayscale and full-to-single input strategies enhance accuracy. Finally, Exp.5 ranks modality importance, with the

⁴SemSeg channels: 1) Floor, 2) Person, 3) Furniture, 4) Others.

⁵Human Parts channels: 1) Background, 2) Head, 3) Torso, 4) Upper arms, 5) Lower arms, 6) Upper legs, 7) Lower legs.

⁶Channels: SemSeg-4, Edge-1, Normals-3, Depth-1, Human Parts-7, Sal-1, Head Pose-3, Face Features (Mouth-1, Eye-1, Arousal-1, Valence-1).

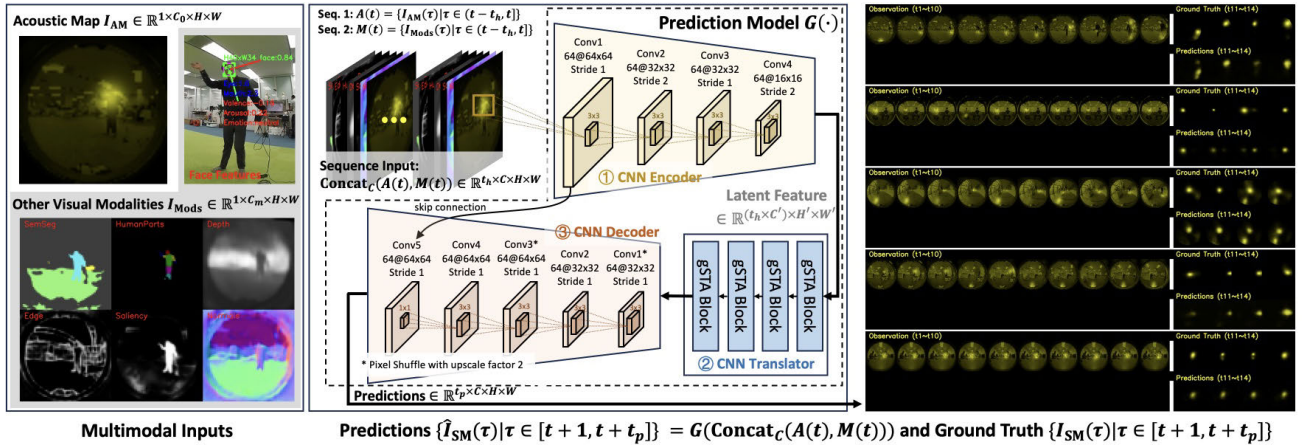


FIGURE 5. Multimodal spatio-temporal prediction model for PSSP using acoustic-visual fusion data.

TABLE 3. Summary of experiments.

Exp.	Title	Objective	Key Findings
1	PSR across prediction steps	Analyze how PSSP accuracy changes over time.	1. Accuracy declines with longer horizons; 2. Proposed Aggregate PSR for later evaluation.
2	PSR Variations	Investigate the effect of kernel sizes and thresholds on PSR.	1. Optimal PSR: kernel size = 5, threshold = 5.0; 2. Fusion ratio of 0.5 is effective; 3. Change in PSR across datasets discussed.
3	Original or fused sound map as output	Compare original vs. fused sound maps as outputs	1. Better accuracy than baseline ^{†1} in MSE and PSR; 2. Original maps as output yield better PSR.
4	SM-only or full-modality prediction	Assess full-to-single vs. full-to-full input-output configurations	1. Grayscale boosts PSSP performance; 2. Fusion ratio impacts vary; 3. Full-to-single is more effective.
5	Supplemental visual modalities	Examine the benefit of additional visual modalities	1. More visual modalities $\neq$ better performance; 2. Modality importance ranked and discussed; 3. Best configuration achieves 68.73% PSR.

^{†1} Baseline: Repeating the final observed original sound map, serving as a reference to assess model performance against static repetition.

optimal configuration achieving a PSR of 68.73%. For more details on the experiments and discussion, see the following subsections.

A. EXPERIMENT 1: PSR ACROSS PREDICTION STEPS

Table 4 reports $PSR_{k5}@5$ of $\hat{I}_{SM} = \{G_{A \rightarrow SM}^{(i)}(A(t))\}_{i=1}^4$ across four prediction steps. PSR declines from $t1$ to $t4$, indicating reduced accuracy over time. To evaluate overall performance, we define Aggregate PSR as the average PSR across all prediction steps. This metric reflects model stability under increasing temporal uncertainty, typically falling between the values at $t2$ and $t3$. A higher Aggregate PSR indicates more consistent peak alignment. For brevity, we refer to Aggregate PSR as PSR in the following experiments.

B. EXPERIMENT 2: PSR VARIATIONS

To identify the most practical evaluation metric, we analyze PSR variations with $\{G_{A \rightarrow SM}^{(i)}\}_{i=1}^4$ across thresholds $\{2.5, 5.0, 10.0\}$ and kernel sizes $\{1, 3, 5, 7, 9\}$. We also assess

TABLE 4. $PSR_{k5}@5$ across the four prediction steps.

SM Ratio	t1	t2	t3	t4	Aggregate PSR
1.0	71.14%	68.09%	66.81%	65.74%	67.94%
0.9	71.47%	68.32%	67.15%	66.14%	68.27%
0.7	71.33%	68.34%	67.18%	66.08%	68.23%
0.5	71.09%	68.31%	67.13%	66.07%	68.15%

^{†1} Models $\{G_{A \rightarrow SM}^{(i)}\}_{i=1}^4$ predict the original sound maps.

^{†2} $t1, t2, t3, t4$ denote the predictions in $+0.5, +1.0, +1.5, +2.0$ s.

the effect of the sound map ratio using four SM-only models with ratios $\{1.0, 0.9, 0.7, 0.5\}$ (Table 5). PSR scores under different settings are reported, with $PSR_{k5}@5$ results per dataset shown in Table 6 alongside baselines, which employ the repeating sound maps of the most recent observation as the prediction. All models use fused sound maps as inputs to predict original sound maps. Based on the results, we draw the following conclusions:

TABLE 5. PSR across different sound map ratios, thresholds and peak distance variations.

Threshold	SM Ratio	Kernel Size				
		$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 9$
2.5	1.0	62.32%	63.13%	63.98%	64.60%	65.15%
	0.9	62.26%	63.32%	64.19%	64.81%	65.31%
	0.7	62.40%	63.46%	64.21%	64.82%	65.36%
	0.5	62.44%	63.46%	64.17%	64.73%	65.24%
5.0	1.0	67.03%	67.43%	67.94%*	68.43%	68.82%
	0.9	67.27%	67.76%	68.27%*	68.75%	69.12%
	0.7	67.36%	67.78%	68.23%*	68.73%	69.08%
	0.5	67.27%	67.73%	68.15%*	68.60%	69.01%
10.0	1.0	71.06%	71.38%	71.93%	72.39%	72.88%
	0.9	71.38%	71.69%	72.26%	72.74%	73.18%
	0.7	71.29%	71.63%	72.22%	72.74%	73.19%
	0.5	71.27%	71.51%	72.08%	72.59%	73.03%

[†] Models $\{G_{A \rightarrow SM}^{(i)}\}_{i=1}^4$ are used.

* Previously presented in Table 4.

TABLE 6. PSR_{k5}@5 of individual datasets and baselines.

SM Ratio	3F	1F	MTG	Chat
1.0	47.26%	84.62%	66.59%	72.94%
0.9	48.12%	84.56%	66.53%	73.58%
0.7	47.70%	84.56%	66.88%	73.82%
0.5	47.12%	84.71%	66.80%	74.21%
Baseline ^{†2}	39.29%	76.80%	58.18%	70.23%

^{†1} Models $\{G_{A \rightarrow SM}^{(i)}\}_{i=1}^4$ are used.

^{†2} PSR for baseline frames is generated by repeating the final observed original sound map without predictions, serving as a reference to assess model performance against static repetition.

1) PRACTICAL IMPLICATIONS OF PSR THRESHOLDS

Over 62% of predictions fall below the 2.5 threshold, with accumulation decelerating at higher thresholds. This indicates a right-skewed peak distance distribution concentrated near zero. In Figure 3, a threshold of 2.5 aligns with the typical 5-pixel human face scale, reflecting highly accurate predictions. A threshold of 5.0 offers a more tolerant criterion, accounting for minor deviations and better matching human perception.

2) PSR RISES WITH LARGER KERNELS

As k raises from 1 to 9, PSR consistently increases. Large kernels blur sound maps. This results in broader regions of high average intensities rather than sharp peaks. Because of this, the peaks are more centralized, resulting in a higher PSR.

3) $K = 5$ AS A PRACTICAL KERNEL SIZE

Kernel size influences model ranking by PSR, reflecting a trade-off between robustness and detail. Larger kernels emphasize the regional structure and reduce sensitivity to minor noise, but may obscure finer details. Given that human faces are the main sound sources, we select $k = 5$ as a balanced choice.

4) PSR DIFFERS ACROSS DATASETS

PSR varies notably between datasets in Table 6. In the fixed talk scenes (MTG, Chat), participants remain visible

TABLE 7. MSE among the ground truth, predictions, and baseline frames.

SM Ratio	Fused Output	MSE	MSE Baseline	MSE with Baseline
1.0	—	0.0055	0.0091	0.0038
0.9		0.0055		0.0038
0.7		0.0055		0.0039
0.5		0.0056		0.0041
0.9	○	0.0046	0.0075	0.0032
0.7		0.0037	0.0060	0.0026
0.5		0.0040	0.0065	0.0032

TABLE 8. PSR_{k5}@5 under the original and fused sound map output.

SM Ratio	Original SM Output*	Fused SM Output Raw	Fused SM Output Transformed	Baseline
1.0	67.94%	—	—	60.68%
0.9	68.27%	68.20%	68.20%	
0.7	68.23%	67.99%	67.95%	
0.5	68.15%	61.12%	62.69%	

* Previously presented in Table 4.

and engaged in conversation, resulting in low prediction errors. In contrast, the mobile noise scenes (1F, 3F) are dynamic environments with fewer pedestrians. The 3F dataset shows higher errors, likely caused by random background noise (e.g., air conditioners, electronics) and off-camera conversations. Meanwhile, the 1F dataset yields relatively lower errors because it is recorded in an empty lobby dominated by motor noise.

5) A FUSION RATIO OF 0.5 IS PROMISING IN TALK SCENES

The overall PSR differences across models are marginal ($\Delta \in [0.18\%, 0.35\%]$). A notable improvement is observed in the Chat dataset, where PSR increases by 1.27% under a lower sound map ratio. This suggests that fused acoustic maps offer additional details beneficial for sound source identification. However, whether lower ratios remain effective in multimodal learning necessitates further investigation, as discussed in Section VI-D.

C. EXPERIMENT 3: ORIGINAL OR FUSED SOUND MAP AS OUTPUT

The output format typically mirrors the input in prediction models [22]. In this study, where fused sound maps are used as inputs, it is unclear whether predicting original or fused maps is more effective. Using fused maps as output presents challenges: their values are harder to interpret, and optimization becomes more difficult due to the need to predict the grayscale component. To address this, we evaluate $\hat{I}_{SM} = \{G_{A \rightarrow SM}^{(i)}(A(t))\}_{i=1}^4$ for original output and $\hat{I}_{AM} = \{G_{A \rightarrow AM}^{(i)}(A(t))\}_{i=5}^7$ for fused output and present the results in Table 7 and Table 8, from which we draw the following conclusions:

1) MODEL OUTPERFORMS BASELINE IN MSE

We report three metrics - (MSE) MSE between the ground truth and model predictions; (MSE Baseline) MSE between the ground truth and baseline frames; (MSE with Baseline)

TABLE 9. PSR_{k5}@5 in SM-only and full-modality prediction.

Mods	SM Ratio	Prediction Type	
		SM-only	Full-Modality
Fused SM* (1 ch)	1.0	67.94%	—
	0.9	68.27%	
	0.7	68.23%	
	0.5	68.15%	
Fused SM & Gray (2 ch)	1.0	68.37%	68.48%
	0.9	68.48%	68.47%
	0.7	68.53%	68.38%
	0.5	68.59%	68.40%
Fused SM & SemSeg & Depth (7 ch)	1.0	68.46%	67.39%
	0.9	68.56%	67.35%
	0.7	68.62%	67.27%
	0.5	68.69%	67.22%

† (1 ch) $\{G_{A \rightarrow SM}^{(i)}\}_{i=1}^4$; (2 ch) SM-only: $\{G_{A+M \rightarrow SM}^{(i)}\}_{i=8}^{11}$ and Full-Modality: $\{G_{A+M \rightarrow SM+Mods}^{(i)}\}_{i=16}^{19}$; (7 ch) SM-only: $\{G_{A+M \rightarrow SM}^{(i)}\}_{i=12}^{15}$ and Full-Modality: $\{G_{A+M \rightarrow SM+Mods}^{(i)}\}_{i=20}^{23}$.

* Previously presented in Table 4.

MSE between model predictions and baseline frames. Across all models, we observe MSE Baseline > MSE > MSE with Baseline. This implies that the model offers more informative predictions than simple repetition.

2) MODEL OUTPERFORMS BASELINE IN PSR

For fused sound map output, we report two PSR metrics calculated from (Raw) direct model output and (Transformed) recovered sound maps by reversing Equation 3 under the assumption that $I_{Gray} = 1$ because future grayscale images are unknown:

$$I_{transform} = \max((I_{prediction} - (1 - \alpha))/\alpha, 0) \quad (9)$$

The recovered sound maps are clipped to 0 to ensure valid values. Across all models and individual datasets, we observe PSR < PSR Baseline, demonstrating superior peak prediction performance compared to static repetition.

3) ORIGINAL SOUND MAP AS OUTPUT YIELDS BETTER PSR

PSR notably drops when fused sound maps are used as outputs, especially at lower fusion ratios. This may result from the model lacking clear guidance on which features to prioritize, as the grayscale component significantly influences fused map predictions. Consequently, sound map values may be suppressed, leading to degraded performance. Moreover, the findings of the transformed version are comparable to those of the raw outputs, highlighting the difficulty in recovering from fused information.

D. EXPERIMENT 4: SM-ONLY OR FULL-MODALITY PREDICTION

In multimodal spatio-temporal prediction, an appropriate input-output configuration is crucial. Two common strategies are: (1) Full-to-Single modality: Using a primary modality A along with auxiliary modalities to predict A ; (2) Full-to-Full modality: Predicting all input modalities. While some studies claim the approach (2) provides richer contextual

information [41], empirical evaluations specific to our task are essential to identify the optimal strategy.

In Table 9, each row corresponds to an experimental setup testing four fusion ratios with additional modalities: (a) grayscale only, or (b) grayscale with SemSeg and Depth. In (a), we investigate whether the grayscale channel adds value despite its prior fusion with sound maps. In (b), we assess the impact of two commonly used visual modalities on model performance. Results are further categorized by prediction type, distinguishing between models trained solely on sound maps (SM-only) and those utilizing all available modalities (Full-Modality), with $\hat{I}_{Mods}$ denotes the prediction of other visual modalities:

- (SM-only) $\hat{I}_{SM} = \{G_{A+M \rightarrow SM}^{(i)}(A(t), M(t))\}_{i \in \{1-4, 8-15\}}$
- (Full-modality) $\{\hat{I}_{SM}, \hat{I}_{Mods}\} = \{G_{A+M \rightarrow SM+Mods}^{(i)}\}_{i=16}^{23}$

1) GRAYSCALE CHANNEL AS A FEASIBLE REFERENCE

Although grayscale information is embedded in fused sound maps, adding a separate grayscale channel improves performance. This benefit arises from disentangling grayscale and sound maps, allowing the model to better leverage the fusion operation. The grayscale channel acts as a reference, helping the model isolate and interpret the acoustic component.

2) SUPPLEMENTAL MODALITIES IMPROVE PERFORMANCE

Incorporating additional visual modalities further improves PSR. These modalities offer rich contextual and structural cues, which help reduce ambiguity in PSSP. A deeper analysis of modality interactions is covered in Section VI-E.

3) SM-ONLY PREDICTION IS MORE EFFECTIVE

SM-only prediction consistently outperforms full-modality output in both settings. This is likely because SM-only output allows the model to focus solely on predicting sound maps, whereas full-modality prediction divides model capacity across multiple tasks, potentially reducing accuracy for the primary target.

4) GRAYSCALE TEXTURE AIDS SOUND MAP LEARNING

PSR change patterns diverge between the two prediction types as the sound map ratio varies.

- **(SM-only prediction)** PSR rises as the sound map ratio gets smaller when a grayscale channel is present. This is because lower sound map ratios introduce more grayscale information, forcing the model to reconstruct clean sound maps from noisy inputs. Furthermore, gray channel provides structural gradients and implicit spatial priors, which are absent in sound maps. Notably, the highest PSR of 68.69% is achieved at a 0.5 sound map ratio with additional SemSeg and Depth modalities. This setup is subsequently utilized in Section VI-E to assess the impact of other visual modalities.
- **(Full-modality prediction)** PSR declines as the sound map ratio drops. In this setup, all input modalities are predicted. At low ratios, the fused input resembles

TABLE 10. $PSR_{k5}@5$ with supplemental visual modalities.

ModSet	mod1 Fused SM (1 ch)	mod2 Gray (1 ch)	mod3 SemSeg (4 ch)	mod4 Depth (1 ch)	mod5 Normals (3 ch)	mod6 Edge (1 ch)	mod7 Human Parts (7 ch)	mod8 Sal (1 ch)	mod9 Head Pose (3 ch)	mod10 Face Features (4 ch)	$PSR_{k5}@5$
ModSet1*	✓	✓									68.59%
ModSet2*	✓	✓	✓	✓							68.69%
ModSet3	✓	✓	✓								68.63%
ModSet4	✓	✓		✓							68.49%
ModSet5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	68.50%
ModSet6	✓	✓	✓	✓	✓						68.57%
ModSet7	✓	✓	✓	✓		✓					68.56%
ModSet8	✓	✓	✓	✓			✓				68.21%
ModSet9	✓	✓	✓	✓				✓			68.58%
ModSet10	✓	✓	✓	✓					✓		68.61%
ModSet11	✓	✓	✓	✓						✓	68.73%
Baseline*											60.68%

† Models $\{G_{A+M \rightarrow SM}^{(i)}\}_{i \in \{8,12,24-32\}}$ outputs the original sound maps in SM-only prediction type.

* Previously presented in Table 8 and Table 9

the grayscale, making feature disentanglement hard. Moreover, the model tends to minimize the easier grayscale loss, resulting in under-refined sound map predictions.

E. EXPERIMENT 5: SUPPLEMENTAL VISUAL MODALITIES

Introducing task-relevant information through multimodal input can enhance model performance. To explore this, we begin by incorporating visual modalities based on ModSet2 with $\hat{I}_{SM} = \{G_{A+M \rightarrow SM}^{(i)}\}_{i=24}^{32}$ in Table 10, the best setup in prior tests with a 68.69% PSR. However, adding more modalities does not guarantee better results due to complex inter-modal interactions. For example, ModSet5 includes all modalities but shows a slight PSR drop of -0.19% compared to ModSet2, suggesting interference among inputs. Likewise, SegSeg in ModSet3 raises PSR by 0.04% over ModSet1, while Depth in ModSet4 reduces it by -0.10% . Interestingly, combining both in ModSet2 yields superior performance, highlighting that carefully selected complementary modalities are more effective than indiscriminate inclusion.

To avoid exhaustive testing of all combinations, we extend ModSet2 by adding only one extra modality at a time, resulting in ModSet6–11. These configurations yield similar PSRs, likely due to the limited extra knowledge provided by new modalities. Nevertheless, the PSR differences reveal the relative contribution of each modality. Based on the comparisons, the ranking of modality importance is: Human Parts < Edge < Normals < Head Pose < Face Features. Among them, ModSet11, which includes facial features, achieves the highest PSR of 68.73%.

VII. DISCUSSION

A. INFLUENCE OF SPEED OF SOUND ON ACOUSTIC MAPPING

The speed of sound in air is given by $c_{air} \approx 331.3 \text{ m/s} + \theta \times 0.606 \text{ (m/s)/}^\circ\text{C}$ [42], where θ is the ambient temperature in

Celsius. This corresponds to speeds ranging from 343.42 m/s at 20°C to 349.48 m/s at 30°C . Given a microphone spacing of 42–178 mm and a sampling rate of 44.1 kHz, the maximum time delay between microphones spans approximately 21.8 samples at 20°C and 21.5 samples at 30°C . This minor variation indicates that ambient temperature has a negligible effect on acoustic mapping. Consequently, we adopt a nominal speed of sound of 345 m/s, corresponding to a typical indoor temperature of 23°C .

B. INFLUENCE OF GRID-MIC DISTANCE ON ACOUSTIC MAPPING

We adopt a fixed grid-to-microphone distance of 1.5 m. To assess its validity, we analyze the geometric delay variation caused by deviations in Grid-Mic distance. A variation of $\pm 1.5 \text{ m}$ can introduce up to 2.3 samples of time difference for off-center sources, leading to phase misalignments that may distort beamforming results. Despite this, the fixed-distance grid generally produces high-fidelity acoustic maps in our scenarios. For improved precision with sources beyond 10 m, adopting an adaptive grid distance may be beneficial.

C. LIMITATIONS AND INSIGHTS

The proposed architecture exhibits moderate accuracy in certain scenarios. A key limitation lies in the limited input resolution, which hinders fine-grained dynamics. The lightweight CNN and moderate data size may underfit complex spatio-temporal patterns, particularly in dynamic settings. Moreover, current evaluation metrics lack perceptual sensitivity, especially for multi-peak cases. These observations highlight the need for higher-resolution inputs, more advanced architectures, and perceptually aligned metrics.

D. SCALABILITY AND FUTURE APPLICATIONS

The current dataset is collected in controlled indoor settings with background and motor noise. Transitioning to unconstrained environments introduces challenges, such

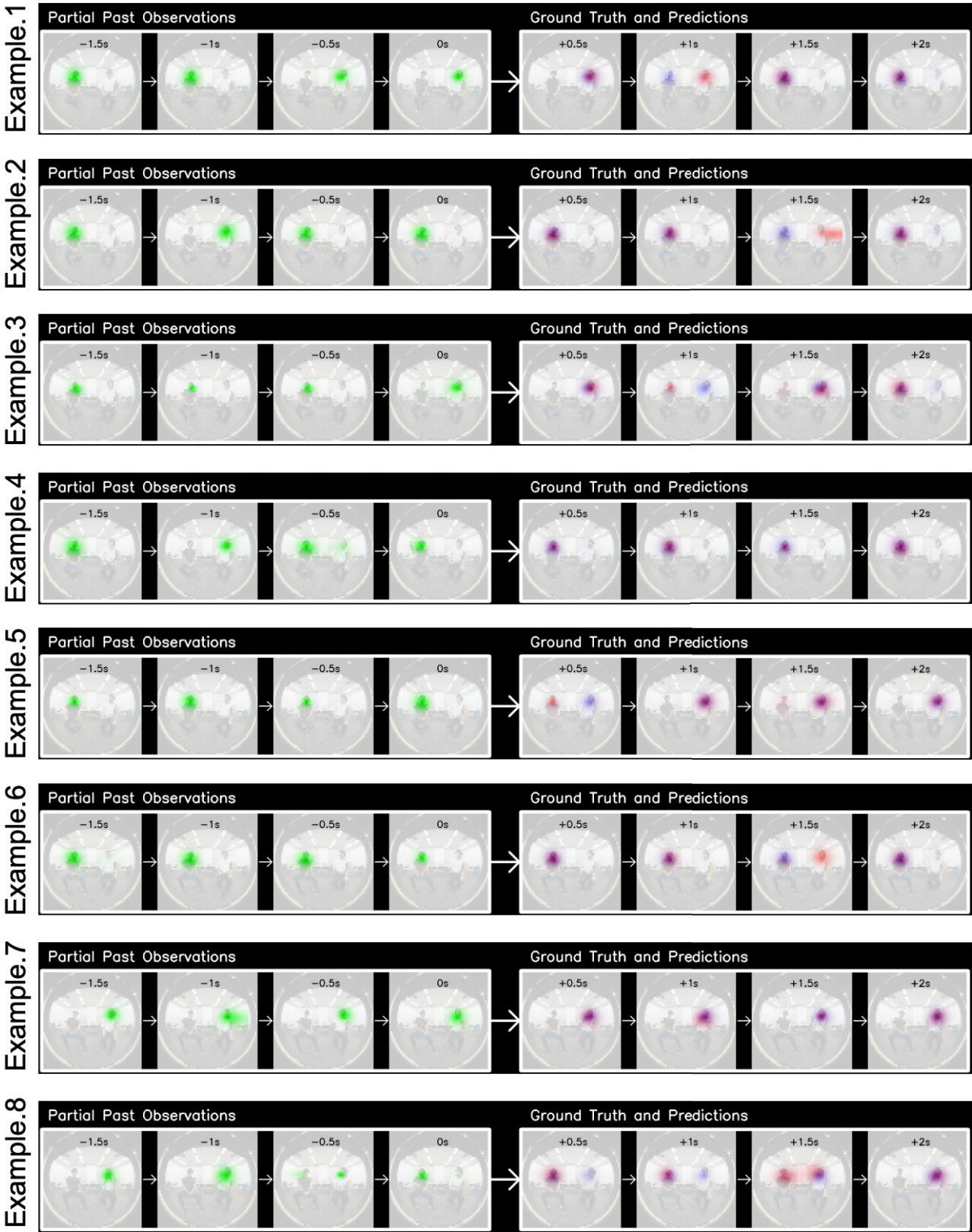


FIGURE 6. PSSP Examples 1–8: Ground Truth vs. Model Prediction of Sound Maps.(Green) Partial observations from the past two seconds;(Red) Ground truth of future sound maps;(Blue) Model predictions of future sound maps;(Purple) Fusion of red (ground truth) and blue (prediction) yields purple, highlighting correctly predicted sound source positions.

as unpredictable sources (e.g., wind). Future work will validate the acoustic-visual sensing system in dynamic, open scenarios, emphasizing robustness and scalability.

Acoustic maps, which transform audio into spatial representations, enable generalization to scenarios with unknown speaker counts. We aim to advance and deploy the PSSP model in real-world multi-speaker HRI. Target applications include managing turn-taking dynamics in elderly care conversations or facilitating group discussions in educational contexts. To support them, integrating acoustic attention and cross-modal alignment will be key to enhancing user experience in such multi-speaker settings.

E. ENHANCED SPEECH RECOGNITION VIA TARGETED BEAMFORMING

By localizing sound sources using acoustic maps, robots can dynamically perform targeted beamforming toward relevant regions. This approach enhances speech clarity, boosting speech recognition accuracy in noisy, multi-speaker environments. Such auditory precision is essential for real-world HRI, fostering context-aware communication and advancing robotic social intelligence.

F. PRETRAINING WITH PSSP FOR DOWNSTREAM HRI TASKS

This study employs a spatio-temporal prediction task, PSSP, and reveals that acoustic maps encode meaningful and discriminative representations. By capturing auditory scene dynamics, PSSP can serve as a pretraining task to extract general-purpose latent features for downstream HRI tasks. Leveraging these informative representations may further enhance the adaptability and reliability of acoustic perception in complex, real-world environments.

VIII. CONCLUSION

This study introduces acoustic mapping as a novel visual modality to enhance robotic perception. By projecting spatial information into 2D maps, robots can access auditory cues beyond verbal content. Through extensive experiments using in-the-wild data, we demonstrate the merits of incorporating acoustic maps into spatio-temporal prediction tasks. Notably, SM-only prediction outperforms certain full-modality configurations, emphasizing the value of task-specific learning. Our results also reveal that more input channels do not guarantee better performance, highlighting the complex interplay among modalities. These findings validate the effectiveness of acoustic maps in interpreting soundscapes and identifying interaction-relevant cues, paving the way for more context-sensitive robotic systems in real-world environments.

APPENDIX A

PSSP EXAMPLES: GROUND TRUTH VERSUS MODEL PREDICTIONS

To illustrate the PSSP task, Figure 6 presents visualizations of eight representative examples, each divided into two

segments. The left segment displays partial observations from the past two seconds (-1.5 s, -1 s, -0.5 s, 0 s). Despite the input spans 10 frames (5 s), only a subset is displayed to enhance clarity. Each frame overlays the sound map onto an RGB image to convey contextual information, with green indicating past sound maps. The right segment depicts model predictions and ground truth over the future two seconds ($+0.5$ s to $+2$ s), with red denotes ground truth, blue represents model predictions. Spatial overlap between red and blue yields purple, indicating accurate localization by the model. Thus, purple regions reflect areas of agreement between predicted and actual future sound source locations. This visualization highlights the model's ability to capture the spatio-temporal dynamics of sound activity in complex, real-world environments.

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ZHICHAO CHEN received the master's degree in engineering from Osaka University, Osaka, Japan, in 2022. He is currently pursuing the Ph.D. degree with the Ishiguro Lab, Osaka University. He was a Technical Staff with the Behavior Learning Research Team at RIKEN, from 2022 to 2023. Since 2023, he has been a Staff Engineer at Panasonic Industry Company Ltd. His research interests include human-robot interaction, android robots, reinforcement learning, inverse reinforcement learning, and multimodal learning.



CHENFEI XU received the master's degree in engineering science from Osaka University, Osaka, Japan, in 2024, where he is currently pursuing the Ph.D. degree. He was a Junior Research Associate with the Guardian Robot Project, RIKEN Information Research and Development and Strategy Headquarters, Kyoto, Japan, since 2024. His research interests include intelligent systems, robotics, and generative models for human-robot interaction.



HUTHAIFA AHMAD received the master's and Ph.D. degrees in engineering from Osaka University, Osaka, Japan, in 2018 and 2021, respectively. Since 2021, he has been a Researcher with the Guardian Robot Project, RIKEN Information Research and Development and Strategy Headquarters, RIKEN, Kyoto, Japan. His research interests include developing robotic systems with body characteristics that allow a safe, natural, and enjoyable physical interaction experience with humans.



YUYA OKADOME received the B.E., M.E., and Ph.D. (Eng.) degrees from Osaka University, Osaka, Japan, in 2011, 2013, and 2016, respectively. In 2016, he joined Hitachi Ltd., as a Researcher, moved to the RIKEN Information Research and Development and Strategy Headquarters, as a Researcher, in 2021, and then moved to Tokyo University of Science, in 2023. He is currently a Junior Associate Professor with the Faculty of Engineering, Tokyo University of Science. His research interests include data analysis, machine learning, robotics, and intelligent systems.



HIROSHI ISHIGURO (Member, IEEE) received the D.Eng. degree in systems engineering from Osaka University, Japan, in 1991. From 1992 to 1994, he was a Research Associate with the Graduate School of Engineering Science, Osaka University. From 1998 to 2000, he was an Associate Professor with the Department of Social Informatics, Kyoto University. From 1998 to 1999, he was also a Visiting Scholar with the University of California, San Diego, CA, USA.

From 2000 to 2001, he was an Associate Professor and a Professor with the Department of Computer and Communication Sciences, Wakayama University, from 2001 to 2002. He then moved to the Department of Adaptive Machine Systems, Graduate School of Engineering, Osaka University, as a Professor, from 2002 to 2009. He has been a Professor with the Department of Systems Innovation, Graduate School of Engineering Science, Osaka University, since 2009, and as a Distinguished Professor with Osaka University, since 2017. From 2002 to 2013, he was the Group Leader of Hiroshi Ishiguro Laboratories, Advanced Telecommunications Research Institute, where he has also been the Visiting Director, since 2014. His research interests include distributed sensor systems, interactive robotics, and android science. Especially, his Android studies are well-known very much in the world. He is currently an ATR Fellow. He received the Osaka Cultural Award, in 2011. In 2015, he received the Prize for Science and Technology (Research Category) by the Minister of Education, Culture, Sports, Science and Technology (MEXT). He was also awarded the Sheikh Mohammed Bin Rashid Al Maktoum Knowledge Award, Dubai, in 2015.



YUTAKA NAKAMURA received the Ph.D. degree from the Graduate School of Information Science, Nara Institute of Science and Technology, in 2004. He was a Postdoctoral Researcher in reinforcement learning for bipedal walking, until 2006. From 2006 to 2020, he was with Osaka University and conducted research on motion generation for robots with complicated bodies and learning mechanisms for communication robots. He is currently the Team Leader of the Guardian

Robot Project, RIKEN, Japan. His goal is to develop robots that learn how to behave in everyday spaces, where they coexist with humans through their own experiences. His research interests include reinforcement learning, motion planning, and generative models for human–robot interaction.

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